Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

Detecting overclaiming relative to guideline-supported evidence

Anonymous ML4H 2026 Findings Track draft

Abstract

Large language models can draft medication alerts whose fluency exceeds the evidence they cite. This paper studies one workflow: pharmacogenomic and genomic medication-alert text. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or parse free text; it tests a deterministic controller over structured annotations. In 30 author-designed synthetic cases, the controller blocked or narrowed all 20 designed overclaim archetypes, allowed all 10 bounded alerts, and produced no inappropriate denials. Action conformance was 30/30; primary-gate conformance was 28/30. These are specification-conformance results, not clinical safety, generalization, or patient-benefit evidence.

1. Introduction

Healthcare LLMs can make evidentiary overreach look syntactically clean. A medication alert may cite a real guideline while turning a bounded pharmacogenomic recommendation into a universal instruction, a population-specific association into a transported claim, or a surrogate endpoint into patient-outcome benefit. The gap matters because fluency can hide what evidence does, and does not, authorize.

Pharmacogenomic medication alerts are a useful testbed because guideline resources such as CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation. That creates a concrete setting for auditing a narrow failure mode: converting real but limited evidence into stronger clinical authority.

2. Related Work and Evidence Boundary

Healthcare AI evaluation should distinguish implementation consistency from patient-outcome benefit. Byrne, Domenico, and Moore emphasize that outcome claims require pragmatic clinical evaluation. Stegenga's Medical Nihilism motivates caution when evidentiary pipelines make interventions appear stronger than they are. FDA-NIH BEST materials provide the endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable.

Pharmacogenomics adds a useful positive control. CPIC-style resources often support bounded prescribing language, such as avoiding a drug in a specific genotype/phenotype context or reviewing dose because of toxicity risk. Those resources do not automatically support claims that an LLM improves outcomes, that an association is causal, that a finding transports across populations without recalibration, or that a weak genetic association authorizes treatment denial.

3. Methods

The controller is a deterministic pre-display gate, not an autonomous clinical agent. In Stage 1, it receives a synthetic candidate alert plus structured evidence annotations assigned in the case bank; it does not parse draft_claim text, call an LLM, verify citations, or extract EHR/population features. The fixed precedence order is citation/guideline support, population fit, then endpoint/actionability. The controller returns one of five permissions: allow, narrow, abstain for population fit, abstain for citation support, or deny unsupported action.

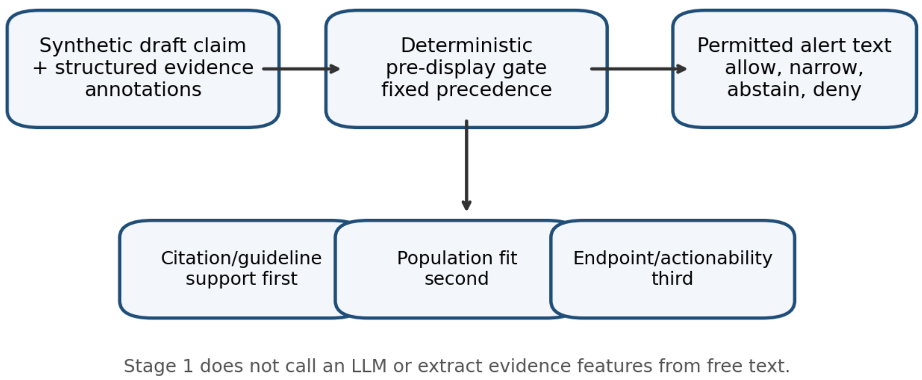

Figure 1. Pruned architecture for the evidence gate.

Gate	Question	Permitted effect
Citation/guideline support	Is the cited guideline real, current, relevant, and strong enough for the claim?	Deny fabricated claims; abstain on uncertain/conflicting support.
Population fit	Does source evidence plausibly fit the target patient/cohort?	Require source-specific validation or abstain from transport.
Endpoint/actionability	Does the evidence support medication action, only surrogate/process language, or no action?	Allow bounded alert, narrow wording, or deny unsupported action.

4. Stage 1 Specification-Conformance Scaffold

The case bank contains 30 author-designed synthetic cases: 10 bounded aligned alerts and 20 overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation weakness, variant uncertainty, or obsolete/context-shifted evidence. The 20/10 split is stress-test coverage, not a clinical base rate. Each case has expected labels, but the controller does not read them.

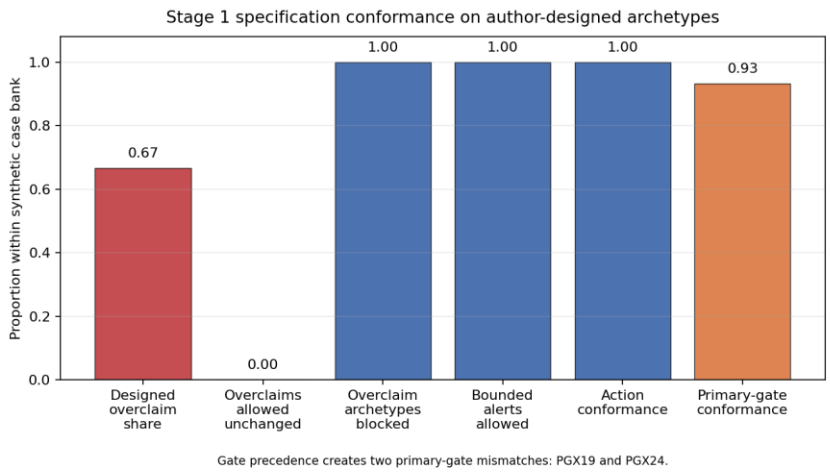

Figure 2. Stage 1 specification-conformance counts on author-designed synthetic archetypes.

Result	Value	Claim allowed
Author-designed overclaim share	20/30	Stress-test case mix, not a measured clinical base rate.
Overclaim archetypes allowed unchanged	0/20	Derived as overclaim cases that reached allow bounded alert unchanged.
Designed overclaim archetypes blocked	20/20	Controller routes intentionally overstrong claims to narrow, abstain, or deny.
Bounded alerts allowed	10/10	Controller permits aligned bounded alerts in the synthetic case bank.
Action conformance	30/30	Actual action matched expected action for all cases.
Primary-gate conformance	28/30	PGX19 and PGX24 differ because citation support fires before endpoint or population fit.
Inappropriate denial	0/10	No aligned bounded alerts were denied too strongly in Stage 1.

Error analysis: PGX19 and PGX24 are action-level successes but primary-gate mismatches. In both cases, citation/guideline support fires before the gate that motivated the case design. This exposes the controller precedence order and prevents the Stage 1 result from being presented as a perfect gate-identification result.

5. Evaluation Plan

The proposed independent evaluation compares ungated LLM drafts with gated drafts on independently authored cases. Reviewers should be blinded to arm when feasible and should classify overclaiming, inappropriate denial, preserved clinician authority, population-fit concerns, citation support, and error category.

- Primary outcome: reduction in unsupported evidence overclaiming.
- Safety outcome: inappropriate denial of bounded, guideline-supported alerts.
- Secondary outcomes: calibration of overclaim risk, interrater agreement, and qualitative error taxonomy.

- Boundary: no real patient data are needed for Stage 1; any Stage 2 use of clinical cases requires appropriate governance or synthetic case authoring.

6. Ethics, Governance, and Limitations

The governance value of the gate is transparency. The controller is deterministic, auditable, and intentionally non-agentic: each alert can show which evidence feature triggered allow, narrow, abstain, or deny. Abstention is a governance action rather than a failure state; it preserves clinician authority when evidence is stale, conflicting, or poorly transported.

Fairness and bias are central in pharmacogenomics. HLA-B*15:02 risk language is ancestry-conditional, founder-variant evidence may not transport to general US populations, and biobank results can misfit underrepresented or recently admixed groups. PGX09, PGX10, PGX25, PGX26, and PGX27 treat population fit as an equity constraint.

PGX24 marks the highest-risk boundary: using a polygenic score to deny opioid or stimulant therapy. That may become algorithmic denial of analgesia or psychiatric care. A defensible alert system should separate metabolism/safety guidance from unsupported addiction-liability claims.

Regulatory placement matters. Stage 1 contains no PHI and makes no diagnosis or treatment recommendation. Deployed clinical decision support would still need accountability for extraction, citation verification, thresholds, overrides, and audit logging.

The key limitation is that results are synthetic and author-designed. They show rule conformance, not generalization, and assume structured evidence annotations rather than extracting them from live LLM output or EHR context.

7. Conclusion

This small deterministic evidence gate separates bounded guideline support from unsupported action, population transport, and citation overreach. It is a Findings-track design and evaluation-plan artifact with a repaired reference implementation.

References and Web Sources

- ML4H 2026 home. https://ml4h.ahli.cc/
- ML4H 2026 call for participation. https://ml4h.ahli.cc/submit/call-for-papers/
- Byrne, D. W., Domenico, H. J., & Moore, R. P. (2024). Artificial intelligence for improved patient outcomes: The pragmatic randomized controlled trial is the secret sauce.. https://kjronline.org/DOIx.php?id=10.3348%2Fkjr.2023.1016
- Stegenga, J. (2018). Medical Nihilism. Oxford University Press.. https://global.oup.com/academic/product/medical-nihilism-9780198747048
- FDA-NIH BEST Resource. https://www.ncbi.nlm.nih.gov/books/NBK326791/
- CPIC guidelines overview. https://cpicpgx.org/guidelines/
- ClinPGx CPIC CYP2C19-clopidogrel guideline. https://www.clinpgx.org/guideline/PA166251443
- CPIC CYP2C19-clopidogrel 2022 update. https://pmc.ncbi.nlm.nih.gov/articles/PMC9287492/
- CPIC DPYD-fluoropyrimidines guideline. https://www.clinpgx.org/guideline/PA166251462
- CPIC DPYD-fluoropyrimidines update. https://pmc.ncbi.nlm.nih.gov/articles/PMC5760397/
- CPIC TPMT/NUDT15-thiopurines update. https://pmc.ncbi.nlm.nih.gov/articles/PMC12997511/
- ClinPGx HLA-B-abacavir guideline. https://www.clinpgx.org/guideline/PA166251444
- ClinPGx HLA-A/HLA-B-carbamazepine guideline. https://www.clinpgx.org/guideline/PA166251448
- ClinVar. https://www.ncbi.nlm.nih.gov/clinvar/
- gnomAD. https://gnomad.broadinstitute.org/
- All of Us Research Program. https://allofus.nih.gov/
- Genome India project. https://genomeindia.in/
- Psifas / Mosaic. https://partnership.psifas.org.il/